# COMPA$_test$

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### Setup

Instantaneous kinematics:

$$\mathbf{v} = J\,\omega,$$

where $\omega$ are the actuator/joint rates and $\mathbf{v}$ is the body/world twist. Assume actuator rates are bounded in a (unit) Euclidean ball:

$$\|\omega\| \leq 1.$$

(If the bound is not spherical, see the "non-spherical bounds" note below.)

### Image of a unit ball under a linear map

The set of achievable twists is the image of that ball under $J$:

$$\mathcal{V} \;=\; \{\,\mathbf{v} = J\,\omega \;:\; \|\omega\| \leq 1\,\}.$$

Linear algebra fact: the image of a ball under $J$ is an ellipsoid whose principal directions are the left singular vectors of $J$ and whose semi-axes are the singular values of $J$.

### SVD viewpoint

You can see this directly with the singular value decomposition (SVD):

$$J = U\Sigma V^\top,$$

so that

$$\mathbf{v} = J\omega = U\Sigma \qquad \underbrace{(V^\top\omega)}_{\mathbf{z},\ \|\mathbf{z}\| = \|\omega\| \leq 1} \qquad .$$

### Quadratic form of the ellipsoid

Among all $\omega$ that map to a given $\mathbf{v}$, the minimum-norm preimage is

$$\omega_* \;=\; J^\top\big(JJ^\top\big)^{-1}\,\mathbf{v}.$$

The squared norm of this preimage is

$$\|\omega_*\|^2 = \mathbf{v}^\top \left(JJ^\top\right)^{-1}\mathbf{v}.$$

Requiring $\|\omega_*\| \leq 1$ gives the ellipsoid inequality

$$\mathbf{v}^\top \left(JJ^\top\right)^{-1}\mathbf{v} \;\leq\; 1$$

which is the defining equation of the *mobility ellipsoid*. Its principal axes are aligned with the left singular vectors of $J$, and the semi-axis lengths are given by the singular values of $J$.

### Interpretation

- Directions corresponding to larger singular values of $J$ are "easier" motions (long axes of the ellipsoid). - Directions with smaller singular values are harder to achieve (short axes). - If $J$ is rank-deficient, $(JJ^\top)^{-1}$ is replaced by the pseudoinverse, and the ellipsoid becomes degenerate (flat) along unachievable directions.